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Fuel atomization in gas turbines: A review of novel technology

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Summary

The atomization of fuel is crucial in the combustion and emission of a gas turbine, and the fuel atomization is continuous without any cycles or strokes. However, to achieve a desired amount of combustion during this continuous process, fuel must be added and mixed with the high-pressure air exiting the compressor in proper proportions. To make the engine as small and lightweight as possible is a constraint and requires the fuel injection, mixing, and combustion to occur within the smallest volume possible. In most cases, this is inefficient and less practicable. A major drawback is the requirement of high injection pressure with relatively small increase in flow rate. In recent years, research was conducted to improve fuel atomization in a gas turbine by using different novel approaches that were simpler, more adaptable, and efficient to enhance the atomization. However, most of these studies were in isolation without any comprehensive literature on recent trends. Therefore, this review attempts to give an insight on recent development of fuel atomization in a gas turbine. Particular emphasis was given to air-, plasma-, ultrasound-, and supercritical fluid–assisted atomization techniques.

KEYWORDS

air-assisted atomization, combustion, fuel atomization, gas turbines, novel technology, plasma-assisted atomization, supercritical fluid–assisted atomization, ultrasound-assisted atomization

1 | INTRODUCTION

Gas turbines are internal combustion engines that are used to generate electricity. A gas turbine consists of an upstream revolving compressor coupled with a downstream turbine and a combustion chamber.[1] All gas turbines generate thrust by providing a change in momentum to the air that enters and leaves the gas turbine.[2-4] The higher the difference in momentum, the greater the thrust that the gas turbine produces.[5]

Liquid fuel in the combustor is burnt in large gas turbine engines, and it is usually pressure-atomized with a high-pressure fuel pump.[6] It is essential to know here that due to reduction in engine dimensions, the high-pressure fuel pump does not level with the engine in terms of size or weight; hence, pressure atomization is considered to be detrimental and damaging in small gas turbines that are usually applied in cruise missiles, smart miniature munition systems, business jets, unmanned aerial vehicles, and a number of other turbine engine applications.[6-8] Simultaneously, small turbine engines can function appropriately at higher rotation rates that might approach 100 000 rpm, and this permits the utilization of rotary centrifugal atomizers so as to work as the fuel spray, substituting the high-pressure fuel pump. In a way, this technique is beneficial and cost-effective,

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simple to use, and lightweight in terms of getting the liquid fuel atomization required for the combustion process.

The gas turbine requires a combustor for combustion to occur; hence, the combustor is a vital component of the gas turbine. Unlike automobiles, gas turbines have a continuous flame inside the combustor, which is lit for as long as the engine is running.[9] Once ignited, the flame is maintained by a constant mixture of fuel to the high-pressure compressed air from the compressor, by using a fuel nozzle. The primary purpose of every fuel nozzle is to atomize the fuel into small droplets to speed up the mixing process of fuel and air.[10] The differences between various fuel nozzle technologies lie in how the droplets are exactly produced. Therefore, the droplet size will affect the fuel atomization effectiveness and efficiency in a gas turbine.

Specifically, the characteristics of an ideal fuel atomizer are listed below.[11-15] Ideally, the atomizer would perform well over the full variety of engine operating conditions and would be cheap, lightweight, simple, robust, and easy to maintain.

These characteristics are including good atomization for an extensive range and quick response to adjustments in fuel flow rates, the absence of flow instabilities, low pumping power requirements, low cost, simple design, lightweight, high accessibility for servicing, and low maintenance. Also, the capability of scaling for design flexibility, little susceptibility to harm throughout manufacture; installation; maintenance; and to gum creation because of heat soakage, design for prevention of carbon deposition at the nozzle face; and internal passage blockage by fuel contaminants. More so, uniform fuel distribution, resistance to oxidation and erosion and to whole equipment vibration and fatigue, thermal insulation for extended part life and fuel protection, and long time between overhauls. Good atomization for an extensive range of fuel flow rates.

Generally, an efficient combustion has gradually become essential due to the rapid increase in commercial gas turbines as well as the consequent rise in atmospheric pollution, which is mostly realized by most general public as exhaust smoke.

There is an increase in the utilization of cost-effective gas turbine engines because industries are pushing towards higher specific fuel consumption and lighter packaging. In addition, this has led to the enhancement in engine operating pressures and temperatures, along with major efforts to decrease gas path losses and enhance component efficiencies. As a result, this trend will continue to exist, and more research and development are required in terms of thermal management of the hot engine unit, comprising the fuel nozzle, combustor, and turbine. Nozzle thermal management, turndown ratio ability, and atomization performance while handling and regulating the correct combustor aerodynamics and low pollutant emissions for the fuel injection system are considered to be the most significant performance features that require optimization. The nozzle wetted-wall temperatures can be reduced, and carbonaceous deposit formation in the fuel delivery passages can be prevented by using multifaceted and expensive heat-shielded designs. Optimization of designs by using current computational methodologies is restricted to competence and its increase in cost. This increased demand of combustor can be fulfilled by using important developments in fuel injection concepts, physical understanding, and computational methods.

1.1 | The need for novel atomization methods

Atomization is the fragmentation of bulk liquid, so it gets into small droplets by using a spray or atomizer.[16] Atomizers are generally categorized into pressure atomizer, pressure swirl atomizer, air-blast atomizer, air-assist atomizers, twin-fluid atomizer, rotary atomizer, ultrasonic atomizers, whistle atomizers, and electrostatic atomizer.[17,18] Different types of atomizers will determine the efficacy of the atomization process, which adversely affects the combustion efficiency in a gas turbine engine.

The atomization of fuel is important in the combustion and emission of a gas turbine. For this system, the combustion is continuous operation[19]; consequently, the fuel atomization in gas turbine is continuous without any strokes or cycles. Nevertheless, to accomplishing the desired amount of combustion during this continuous process, fuel must be added and mixed with the high-pressure air exiting the compressor in proper proportions. The constraint to design the turbine engine as lightweight and small as possible will require the different parts of the engine, such as injection, mixing, and combustion of the fuel place to be within the smallest volume possible. This is inefficient and, in most cases, less practicable. Furthermore, in the case of pressure atomizers, a major drawback is the requirement of high injection pressure with relatively small increase in the flow rate; for example, for simplex nozzle, the flow rate varies as the square root of the injection pressure differential ΔPF. Therefore, there is a need for a nonpressurize alternative means of atomization.

Generally, acceptable atomization improves the complete combustion and mixing in a direct injection engine, and consequently, it is an important factor in efficiency and engine emission. In the case of biodiesel, which exhibits difficulty during a cold start due to its

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crystallizing property at low temperatures, the need for atomization as an option to overcome some of these challenges cannot be overemphasized. The probability of biodiesel to be a renewable fossil fuel standby for gas turbine operations is currently being researched on due to earlier reports on some oxides of sulfur (SOx), carbon monoxide (CO), and oxides of nitrogen (NO_x) emission levels. From the environmental point of view, other fuels like biodiesel are much preferred to be used in combustion systems. However, they have yet to be fully optimized for emissions; consequently, the probability of biodiesel to be a low emission alternative fuel option is still at the beginning and being evaluated. With improved atomization, gas turbine operations can realize improved emissions as compared with those that use conventional diesel.[20]

Most atomization techniques, though with many merits, have shown inadequacies in fuel atomization in gas turbines.[21-23] This is due to relative increase in dynamic viscosity and surface tension. The fluid properties are heavily tied to atomization behaviour, whereby the increased viscosity and surface tension limit droplet breakup will lead to larger average droplet sizes, thus increase residence time and NO_x formation.

Although there are other modes of improving emissions, improved fuel injector design and mixing were shown to reduce NO_x, CO, and HC outputs.[24-26] Among the noted efforts to achieve improved atomization is the use of fuel additive.[27,28] Javed et al[29] demonstrated that emissions from a direct injection diesel engine can be decreased by adding metallic fuel-borne additives (zinc oxide). However, the use of additive for atomization increased hydrocarbon (HC) emission with larger particle size and also increased smoke opacity as compared with conventional method.

Other researchers have tried to improve the atomization of fuel using different designs of atomizers. Mlkvik et al[30] designed four different types of internal-mixing twin-fluid atomizers and compared their performance with two well-known designs, namely, the Y-jet and OIG "Gutside in Gas" effervescent atomizers. In their study, they focused on spray stability, primary breakup, internal liquid-gas flow, and droplet size. The results of their studies showed that depending on the gas-to-liquid ratios, the flow rate was enhanced, leading to improved combustion. However, this method is far from efficient because it depends on complex designs and cannot be used in most engines.

Moreover, any variation in the pressure to produce the supply liquid or area of the nozzle outlet will result in the influence of all reported modes of atomization enhancement, either through pressure burners or spray heads. Nevertheless, the technique that was previously mentioned can result in poor liquid atomization at a low flow rate (under low pressure). It can serve as a cure where air or steam can also be utilized on medium- or large-sized turbines so as to facilitate the liquid atomization, requiring more complex and expanded apparatus. Conversely, if the latter method was focused, an extremely complicated construction of the nozzle is needed, which is unmanageable and hard to control. To overcome the drawbacks of the above prior art, a more simple, adaptable, and efficient way of enhancing atomization will become imperative. Attempts were made to impart novel technologies, such as ultrasonic waves to liquid material, by injecting the jet through the injection nozzle under pressure. However, most of these studies were carried out in isolation with no comprehensive reports on novel advances approach made for fuel atomization in gas turbine operations. Therefore, more research is required in order to develop novel atomization designs to accommodate and optimize the of gas turbine performance. Therefore, the main objective of this review is to give an insight on recent fuel atomization developments in gas turbines. Therefore, to accomplish this, particular emphasis is given to air-assisted atomization, plasma-assisted atomization, ultrasound-assisted atomization, and supercritical fluid–assisted atomization.

2 | APPLICATION OF ATOMIZATION IN GAS TURBINE OPERATIONS

Rachner et al[31] concentrated on the utilization of gas turbine atomization in jet condition. In their review, they explored the infiltration and atomization of a plain lamp oil fuel jet in air crossflow and contrasted the trial at test conditions that were significant to lean premixed prevapourized burning in gas turbines. Distinctive procedures were connected, incorporating time-settled shadowgraphs, Mie-dispersing laser light sheets, and phase-Doppler-anemometry; thus, subjective data on jet breaking down and also quantitative information with respect to jet infiltration, fuel situation, and drop size conveyance were acquired. According to Hashimoto et al,[32] the important combustion features of palm methyl ester (PME) can serve as substitute fuel for gas turbines. Their study conducted the chemical equilibrium calculations and studies of fuel atomizing characteristics by utilizing a laser diffraction spray analyser. The study observed a similarity between PME combustion characteristics and that of diesel fuel. Also, it was directed that the utilization of PME instead of diesel fuel for gas turbines can result in the NO_x emissions reduction. Lasheras and Hopfinger[33] analysed the liquid jet instability and atomization in a coaxial gas stream. The mentioned procedure included

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periodic stripping of liquid sheets or ligaments, which then disintegrated into smaller lumps or drops. Models to forecast the liquid shedding frequency, along with the global factors of the spray like spray spreading angle and the liquid core length, were explained in detail. Jiang et al[34] concentrated on jet streams in atomization and showers of gas turbine. They inspected the physical displaying of atomization and showers with regard to computational fluid dynamics, with demonstrated suspicions and confinements examined. Likewise, they talked about the numerical techniques in cutting-edge computational fluid dynamics of atomization and showers, including high-arrange numerical plans. More details are shown in Table 1.

3 | AIR-ASSISTED ATOMIZATION

The air-supported spray density, ethanol atomization characterization, and a variety of biodiesels were studied by Kourmatzis et al.[39] They specified that an air-assisted spray was treated with phase Doppler anemometry or laser Doppler anemometry as well as high-speed microscopic imaging by utilizing different biodiesels: ethanol, medium length, and long chain length, fatty acid methyl esters of short as depicted in Figure 1. The droplet size and momentum decay features of the four fuels were matched as a task of mass loading, Reynolds number, and radial position for several downstream settings. It was recommended by the phase Doppler anemometry or laser Doppler anemometry findings that the sprig characteristics were very comparable past $x/D = 5$, depicting that a greater number of disintegration occurred in the near exit plane region, with insignificant secondary atomization happening downstream.

The twin-fluid or two-phase or pneumatic atomization is basically the atomization of a liquid produced through a compressible fluid, such as gas, steam, or air. Eventually, the techniques and atomized droplets to produce finely atomized sprays are required in a number of industrial procedures. This is the reason for the development of new types of atomizer the recent times. To achieve repeatable results, sophisticated process methods have enhanced the requirement for a precise and single definition in regard to the features of a given spray. As a matter of fact, parameters that describe a given spray are well defined in Table 2 and design engineers can use them at any moment.

The influence of atomization gas properties was analysed by Aftel[40] on droplet atomization in an air-assisted atomizer. To define the impact of these gases based on number density, mean droplet size, distributions in kerosene fuel sprays, velocity, and spray flames, nitrogen, air, carbon dioxide (CO_2), and argon were used as the atomizing gas in an air-assisted spray nozzle by simply utilizing a 2D-phase Doppler interferometer. The atomizer with different gases determined the essential differences in atomization characteristics. In addition, this determination was done under circumstances of momentum flux and constant gas mass. The results also illustrated that the presence of oxygen in the air atomized spray facilitated in the combustion process due to the fact that the atomized spay was small and quick in pace droplets, particularly at locations that were close enough to nozzle exit.

A fuel atomizer (liquid siphon type) facilitated by air was developed by Watanawanyoo et al[10] for a continuous combustor. Refined palm oil and automotive diesel oil that were used as fuels for the atomization scheme in the experiment were used. Likewise, the atomizer was designed by keeping in view the air flow so that the air could flow over the small nozzle. Moreover, stainless steel was used for manufacturing a combustor chamber so as to prevent oxidation at high combustion temperature. As for the outcomes, practical combustion performance by using refined palm oil as a fuel with ultralow CO and HC, emissions were less than 206 and 7 ppm, respectively. Table 3 demonstrates the characteristics of the air-assisted atomization.

According to previous studies, the potential feasibility of utilizing the air-assisted atomization was highlighted in operating gas turbines. This method was shown to be more superior to the conventional methods. The air-assisted atomization was found to improve droplet size and momentum decay to produce finely atomized sprays. This improvement resulted in the droplet size to range from 17 to 200 μm by using a twin-fluid atomizer.

4 | PLASMA-ASSISTED ATOMIZATION

Worldwide, many initiatives were taken with the intention of enhancing the combustion procedures of fossil fuels in heat engines and propulsion systems. If the trending attitudes are focused in this regard, several electrical discharges and plasma were employed in research to accelerate and enhance ignition and combustion processes.[37] Matveev et al[44] examined the plasma fuel nozzle (PFN) concept and stated that it is the most potential tool to understand plasma-assisted combustion that occurs in different heat engines (Figure 2). The outcomes of plasma source choice and optimization, fuel atomizer options deliberation and comparison, and impact on the combustion process parameters were stated. The aforementioned conclusions

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TABLE 1 Different applications of atomization in gas turbine operations

Type	Atomization Conditions	Measurement Techniques	Fuel Type	Advantages	Disadvantages	Reference
Plain liquid fuel jet	Air velocity: 100 and 75 m/s, Air pressure: 6 and 9 bar, Nozzle diameter: 0.45 mm, and Liquid-to-air momentum flux ratio: 2 and 18	Mie-scattering laser light sheet, time-resolved shadowgraphs, acceleration wave modes, and phase-Doppler-anemometry	Kerosene fuel	The atomization computing time in this type is 1 min on a usual 400-MHz PC.	It is not applicable for coaxial injection or injection with very high liquid delivery or injection into still air pressure.	Rachner et al[31]
Liquid jet in gas stream	Dynamic pressure: 180 lb/ft^2	Capillary and	—	This type provides high atomization gas velocity	It is complex technique	Adelberg[35]
Liquid jets	Velocity: 3.5 to 11 m/s Nozzle diameter: 1.4 mm Nozzle length: 150 mm All experiments were conducted under atmospheric pressure.	Liquid sheet flow test	—	This type predicted the formation conditions and shape of an upwash through theoretical analysis.	More studies ought to conduct in order to provide experimental results.	Inamura[36]
Gas turbine	The temperatures of palm methyl ester fuel is 51.85°C. The temperatures of diesel fuel is 29.85°C. Kinematic viscosity is 3.29 mm^2/s. The measurement plane of the LDSA is 25 mm below the nozzle. The atomizing nozzle is 0.4 mm. Swirl chamber is 1.37 mm. The atomizer constant is 0.33. Fuel atomizing pressure is 0.69 MPa. Excess air ratio: 2.53 Fuel flow rate is 1.16 g/s for PME fuel. Fuel flow rate is 1.13 g/s for diesel fuel.	Laser diffraction spray analyser	Palm methyl ester and diesel fuel	The equilibrium calculations of the atomizing characteristics found better results by utilization palm methyl ester as an alternative fuel as compared with diesel fuel in operation gas turbines.	Many parameters and considerations have to be set for operating this technique	Hashimoto et al[32]
Turbulent liquid jet	Liquid to gas density ratio: 10 Momentum flux ratio 6.6, Nozzle diameter: 1.3 mm Jet velocity: 97.84 m/s	Refined level set grid method	Gaseous fuel	Provide significant results of atomization drop size and velocity distributions in operating gas turbines	It is a highly complex process	Herrmann[37]

(Continues)

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TABLE 1 (Continued)

Type	Atomization Conditions	Measurement Techniques	Fuel Type	Advantages	Disadvantages	Reference
Turbulent jet	The annular flow with a 150-mm diameter A short tapered contour nozzle of 10-mm inner diameter The fuel was a mixture of hydrogen with nitrogen with a volumetric ratio of 3.5. Flow velocities for the fuel is 3.55 m/s. Flow velocities for the air jets is 37.9 cm/s. Frequency of 14.8 Hz	Two-phase reacting flows and laser sheet	Mixture hydrogen with nitrogen fuel	Provided better predictions from the simulations Improved the findings of experimental and theoretical	Required to define many parameters in order to apply this technique	Grisch et al[38]

Abbreviation: LDSA, laser diffraction spray analyser.

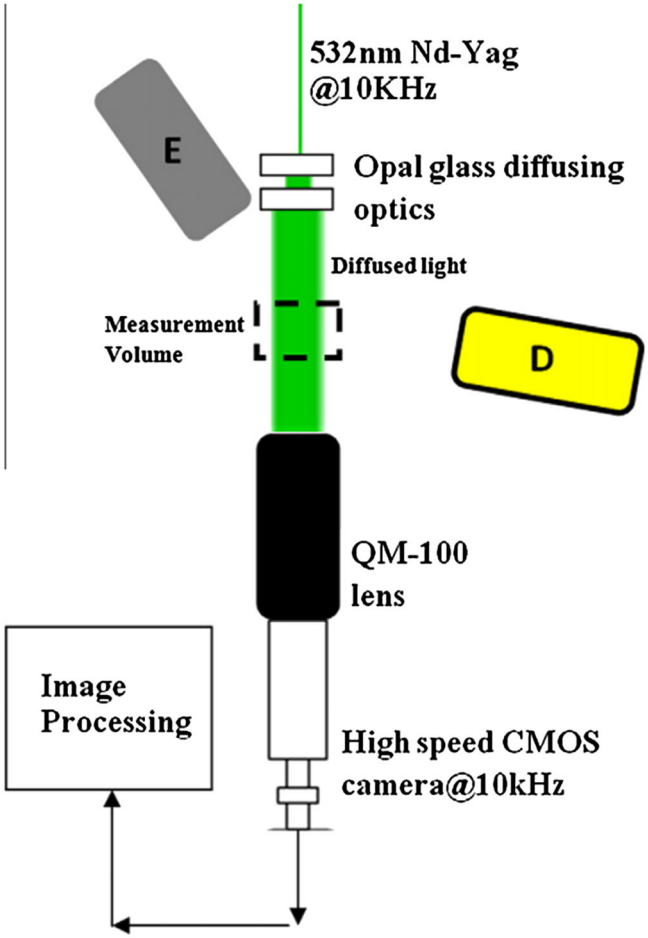

FIGURE 1 Diagram of phase Doppler anemometry or laser Doppler anemometry and high-speed microscopic imaging system experimental measurement techniques, where E is the emitter and D is the detector of the phase Doppler anemometry or laser Doppler anemometry system (the two techniques were not conducted concurrently). Source: Kourmatzis et al[39] [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 The parameters of spray

Arithmetic Mean Diameter AMD (D10)	AMD measured the total number diameters of the drops in the sample spray.
Volume mean diameter VMD (D30)	VMD is the drop diameter, which has the mean arithmetic from the total number of drops in sample spray.
Sauter mean diameter SMD (D32)	SMD is the drop diameter, which has volume/surface ratio arithmetic mean from the total number of the drops in the sample spray.

illustrated that PFN was the most progressive and useful solution for lean burn and low Btu fuel combustors, STIG cycle turbines, liquid and gaseous feedstock gasification unit, and flexi-fuel systems (Figure 3).

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TABLE 3 The characteristics of air assist atomization

Atomizer Type	Gas Type	Atomization Conditions	Measurement Techniques	Significant Results	Reference
Multiple stage atomizer	Ethanol and range of biodiesels (medium and long chain length, fatty acid methyl esters of short)	Density (kg/m^3) 877 to 789, Viscosity (Pas) 0.00199 to 0.0013, Surface tension (N/m) 0.025 to 0.024	PDA/LDA and high-speed microscopic imaging	Droplet size characteristics as well as momentum decay produce similar results of three biodiesels and ethanol past $x/D = 5$.	Kourmatzis et al[39]
Coaxial gas stream	Hydrogen gas	Velocity: 0.42 m/s, Weber number: 38, 58, 118, Density ratio; 1 to 10–3	Reynolds-Weber number parameter space	The current tests of radial variation for droplet size found that more swirl can affect significantly on the drop size at the periphery of the spray, however, it reduces the mean drop size at the centre.	Lasheras and Hopfinger[33]
Twin-fluid atomizer	—	Air flow rate: 0.2 to 2.7, Air pressure: 68.9 to 689 kPa, Air mass flow rate: 0.0004 kg/s to 0.0086 kg/s.	Shadowgraph technique Characteristics of near nozzle flow.	Various sizes ranging from 17–200 μm where found from a twin-fluid atomizer for droplet size.	Watanawanyoo et al[10]
Solid cone sprays and flat sprays	Crude glycerine, Natural gas, Hydrogen	Pressure: 22 mbar, Flow rate: 40 m^3/h, Dynamic viscosity (N s/m^2) at 80°C 22.2.	Scanning electron microscope X-ray detector (energy-dispersive X-ray spectrometer)	The atomizing air can make the spray fineness of the two atomizers increases to glycerine mass ratio up to 1.	Queirós et al[41]
Twin-fluid	Ethanol	Liquid orifice areas: 0.40, 1.82 and 5.07, Liquid flow rates (cc/s) equivalent to 30- and 50-kW energy input, Viscosity: 1.074 cP, Density: 789 kg/m^3.	The camera resolution and focus determined the velocities and droplet size measurements.	Bio-oil/ethanol mixtures containing can reach up to 40% bio-oil if it has low water content (12.60%).	Lujaji et al[42]
Twin fluid		Air diameter: 15 mm, Droplets with Sauter mean diameter in the 15- to 85-μm range, The dynamic range: 2 to 800 μm.	Malvern particle size analyser include photo optic detector and laser source	Spray characteristics separated from each other through simultaneously altering the supply pressure of the atomizing air as well as the liquid.	Lal et al[43]

Abbreviation: PDA/LDA, phase Doppler anemometry or laser Doppler anemometry.

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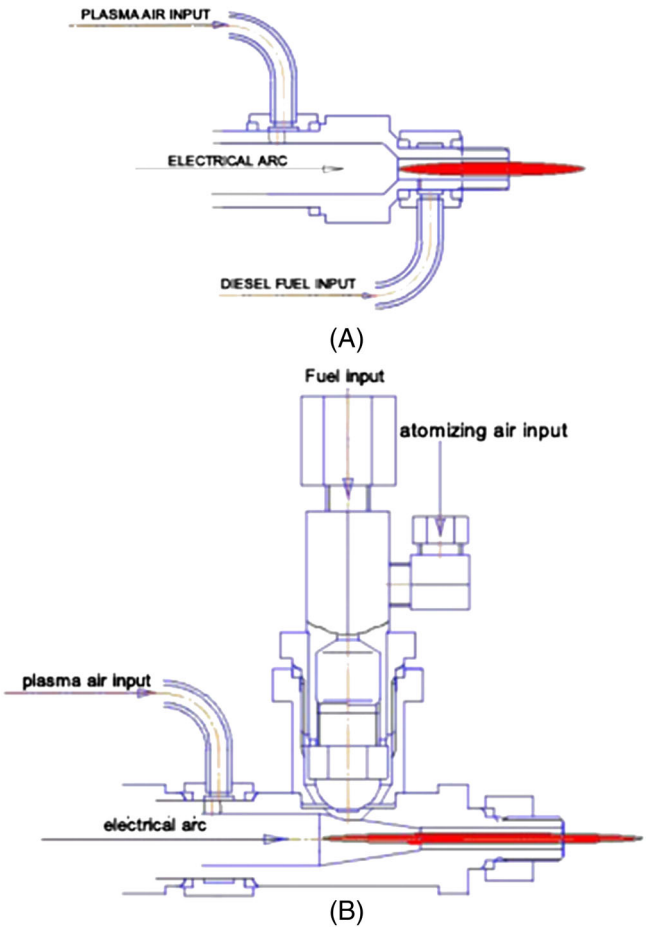

FIGURE 2 Direct fuel feeding into the arc chamber (A) fuel was fed through the 0.3-mm orifice, and (B) fuel was fed through the Hago 609-11 standard nozzle. Source: Matveev et al[44] [Colour figure can be viewed at wileyonlinelibrary.com]

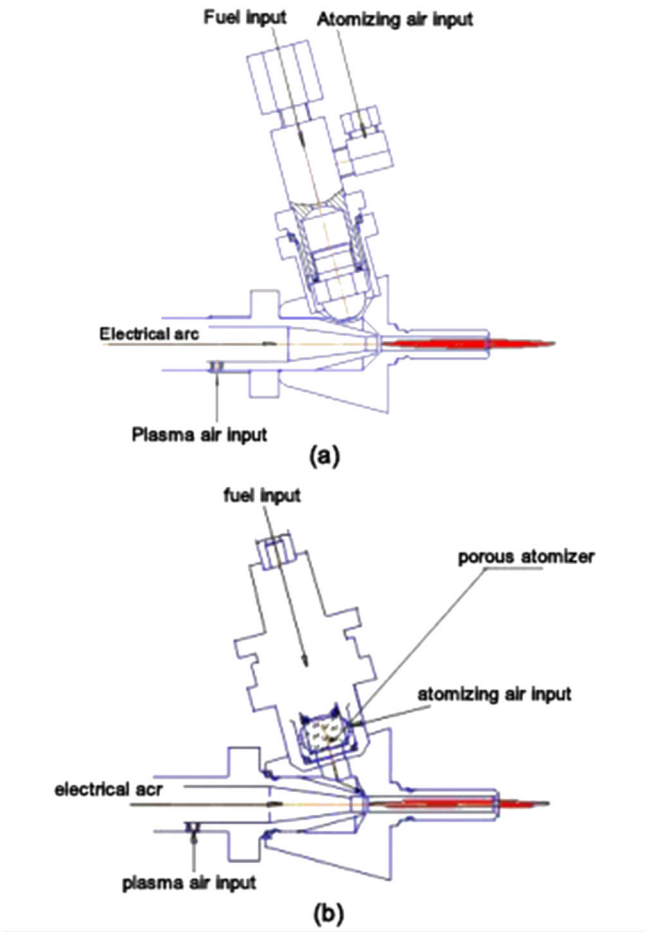

FIGURE 3 Plasma fuel nozzle designs with conventional centrifugal Hago atomizer (A) and porous element (B). Source: Matveev et al[44] [Colour figure can be viewed at wileyonlinelibrary.com]

The on-board plasma-assisted conversion of heavy HCs in the form of synthesis gas was studied by Gallagher et al.[45] They employed *n*-tetradecane (C14H30) fuel, which was improved with plasma under partial oxidation circumstances in the absence of metal catalysts and steam. Hydrogen yield and energy conversion efficiency were the two factors used to compare the performance of all involved components. It was evident from the results that gliding arc systems had the potential to reform heavy HC fuels with high conversion efficiency. The gliding arcs are significant tools of technology for on-board vehicular reforming systems that need more developments.

A new life was offered to the PFN concept by the current developments in microelectronics and growth of nonthermal plasma torch with non–steady-state gas discharge.[46] Nonthermal plasma has a number of benefits like higher stability, longer lifetime, lower arc temperature, and lower averaged power consumption owing to the nature of the non–steady-state discharge, chances of direct fuel injection into the arc chamber, wider operation pressure, and flow range and unique portable design. These benefits are considered responsible for the new efforts made in terms of PFN development for power generation and propulsion. Similarly, a small-scale modelling of the suspension plasma spraying mechanism was explained by Girard et al.[47] The heat transfer that occurred between a droplet of pure water and an Ar/H2 plasma jet was taken into account and investigated. In the flow, low settling time of droplet before affecting the substrate resulted in consideration of radiation as the minor mechanism instead of the major one, whereas convection and conduction were developed further as the droplet lost its shape.

Leblanc et al[48] investigated the transmission electron microscopy of lithiation of spherical silicon (Si) nanopowder produced by induced plasma atomization. The properties of nanopowder were characterized by physical and electrochemical methods. The result found that the primary particles were crystalline with spherical shape, and the diameter measurement ranged from 50 to 200 nm (Figure 4A). Figure 4B shows the transmission electron microscopy micrograph of the crystalline lattice

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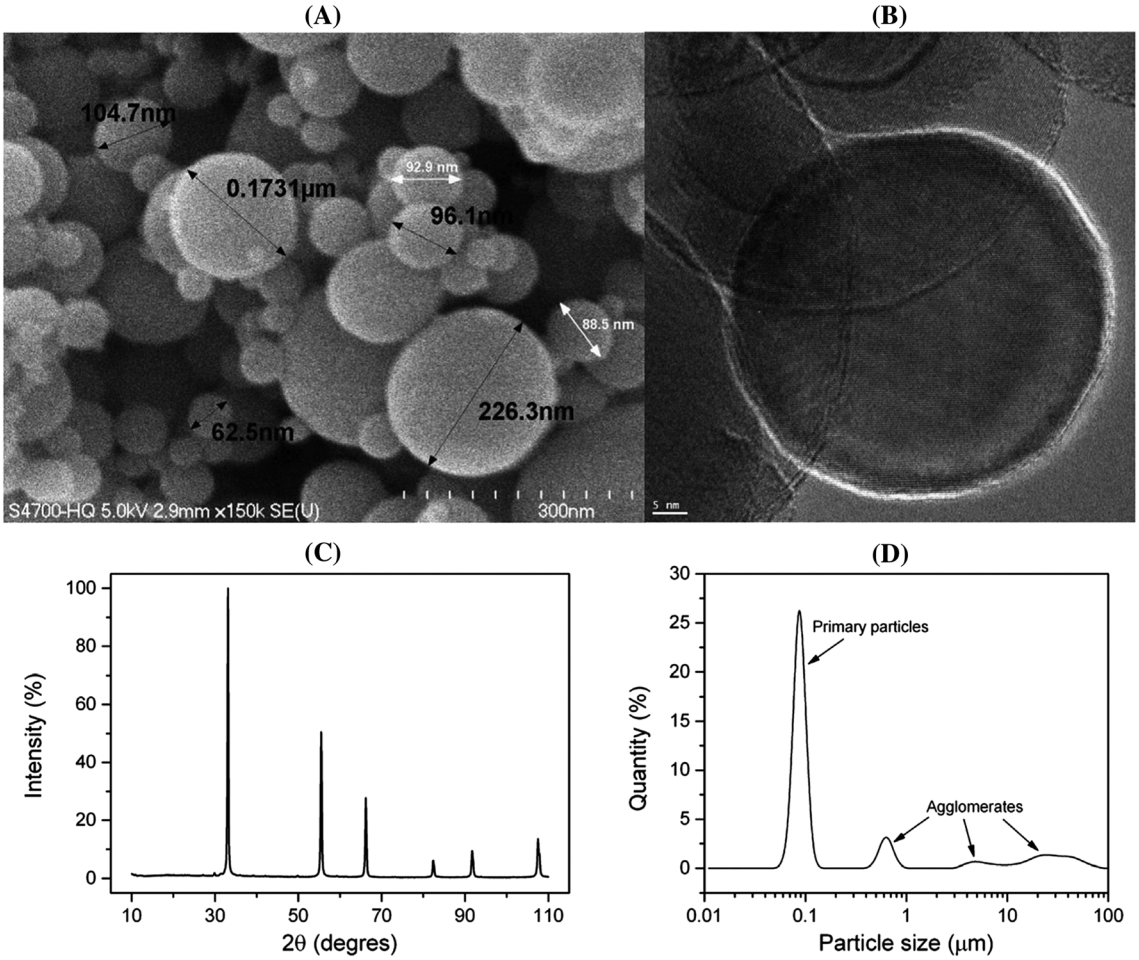

FIGURE 4 (A) Scanning electron microscope micrograph, (B) transmission electron microscopy micrograph, (C) X-ray diffraction diffractogram, and (D) particle size distribution of the silicon nanopowder. Source: Leblanc et al[48]

in the Si nanoparticles, which have a contamination-free atomic surface. In addition, Figure 4C demonstrates the X-ray diffraction diffractogram of the nanopowder, along with the clear identification of Si with the diamond cubic lattice without any impurity phase. Moreover, Figure 4D shows the particle size distribution and validates that the c-Si primary particles formed by plasma procedure have a particle size between 50 and 200 nm with a few greater agglomerates. Furthermore, the Si nanopowder demonstrated a high gravimetric capacity (4900 mAh/g) at first discharge and about 12% irreversible loss of lithium.

On the basis of literature review, plasma-assisted atomization was found to be very important in gas turbines. This technique was superior over conventional techniques in terms of having a controlled injection of much finer particles than in the conventional thermal spraying. Therefore, plasma-assisted atomization allows to achieve thin and finely droplet in fuels between a few tens of micrometres and a few hundreds of micrometres. As a result, operating gas turbines with plasma atomization have improved in combustion efficiency as compared with those of conventional thermal spraying.

5 | ULTRASOUND-ASSISTED ATOMIZATION

In a number of industrial procedures like spray drying, film coating, preparation of fine powders, making emulsion spray cooling, incineration and liquid fuel, and waste combustion. Liquid atomization is of significant importance.[49] Most of these applications require droplets of required size distribution. If several kinds of atomization processes were focused, many of such processes would have been practiced, which can be categorized by keeping in view the mode of energy transfer to reach the liquid sheet surface. Mechanical energy is utilized by mechanical or conventional atomization processes, for instance, two-fluid atomization, pressure atomization, and spinning disk atomization, so as to put pressure on the liquid sheet and enhance its kinetic energy for potential breakdown in the form of droplets. If the energy required by such disintegration processes was compared with the actual energy required in terms of theoretical framework, the amount or value in the processes will be higher. In these processes, there is no control on the final droplet

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size and the velocity with which the droplet was expelled. Therefore, newer processes must be developed so that a desired droplet size distribution is achieved with efficiency. Ultrasonic irradiation for inducing atomization is one of the major techniques to get a narrow droplet size distribution. This is different from the conventional atomization in a way that it can be more energy efficient and only requires electrical energy to be transferred to a piezoelectric vibrating disk. The supplied electric energy results in mechanical vibrations and is used for the generation of droplets. In addition, the ejection of fine droplets from a liquid film made on an ultrasonically vibrating surface is known as ultrasonic atomization. The following two hypotheses fully explain the ejection of the droplets from a vibrating surface:

- Capillary wave hypothesis: This hypothesis can be visually observed, and it takes into account the creation of capillary waves made up of crests and troughs on the vibrating surface.
- Cavitation hypothesis: This hypothesis signifies that cavitation handles and regulates the droplet formation. Cavitation is a process in which cavity formation takes place in the liquid film on the vibrating surface of an atomizer. Bubbles are formed and then disintegrated, which leads to the direct ejection of the droplets, and thus, the velocity of ejection will be considerably higher, owing to the kinetic energy.

To understand the impact of several equipment parameters, such as operating frequency, power dissipation, and the operating parameters like the flow rate and liquid features on the droplet size distribution, Ramisetty et al[49] investigated the measurements of droplet size distribution in an ultrasonic atomizer by utilizing the photographic analysis. By employing photographic analysis, based on the capture of the growth of instability and sudden ejection of droplets with high velocity, mechanistic information in regard to the atomization phenomena was achieved.

- It was evident from the conclusion of the study that an increase in the frequency of atomizer resulted in the decrease of droplet size.
- An increase in the intensity of ultrasound also caused the droplet size distribution to change from a narrow to wider range.
- An increase in the fluid viscosity decreases the drop size.

The findings made the entire process of operating parameters selection for getting an anticipated droplet size distribution by utilizing ultrasonic atomization. Also, this can be helpful to determine the controlling mechanisms for the creation of droplet. An empirical relationship was also formed on the basis of the liquid and equipment operating features for the droplet size prediction.

A relationship determined on the basis of dimensionless numbers like the Weber number and the Ohnesorge number was suggested by Rajan and Pandit[50] to explain the influence of different physicochemical characteristics like the surface tension, viscosity, and density of the liquid on the concluding droplet size. Weber number is basically the ratio between the aerodynamic forces and the surface tension forces, while Ohnesorge number is the ratio between the internal liquid viscosity to surface force and vibrational amplitude. The dimensionless numbers were also altered by introducing the ultrasonic vibrational conditional variables to explain the influence of ultrasound-induced cavitation.

If we focus on the structure of an ultrasonic atomizer, it has a vibrating plate with a concentric hole so as to facilitate the spread of the introduced liquid (Figure 5). This vibrating plate is linked to the transducer, which gets energy from an electric generator. This generator varies the amount of power offered to the atomizer tip by changing and adjusting the percentage amplitude of the output power from the generator. A peristaltic pump is utilized to deliver the liquid on the atomizer tip through the concentric hole. The droplet size and velocity of ultrasonically atomized liquid were measured by using the imaging technique.

Ultrasonic atomizers function on a very refined process, which has the following two steps:

- In the first step, tiny water jets are inserted into a high-speed air flow, which further offers a first disintegration and atomization of the fluid.
- If we focus on the second step, it further has two functions in the form of two phase flow. In this step, the air entering liquid droplets experiences a field of sound waves, which further results in disintegration and a lower droplet measurement. This is experienced through the impact between the two phase flow and a resonator placed in front of the nozzle orifice.

The only way to produce ultrasonic atomizers is through high-precision machining operations. This offers amazing benefits, such as the droplets in the atomized jet demonstrate low values for the SMD along with a narrow range of individual droplet diameter. It can be stated that the drops are quite small and there existed a little difference between the smallest and biggest droplet diameters. This showed that the spray was made up of very small and similar in size droplets, which were significant factors that were realized in all evaporative processes, for

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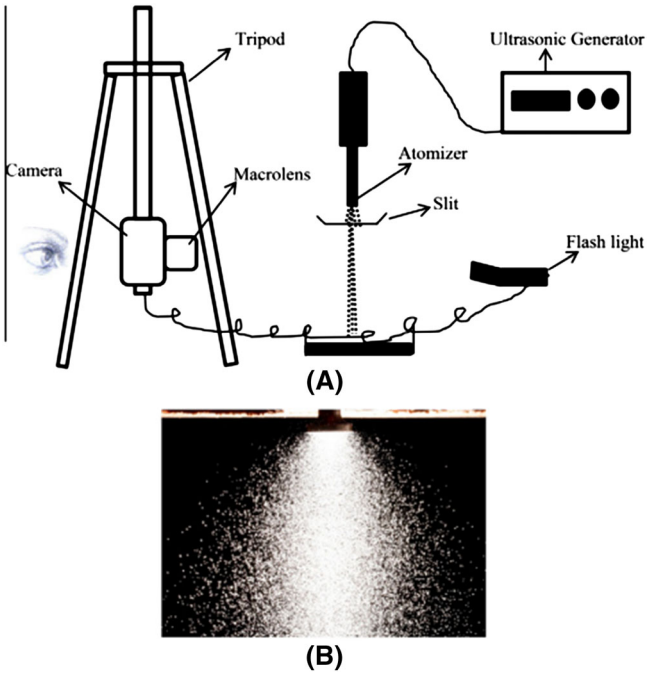

FIGURE 5 (A) Schematic representation of the experimental setup (B) a representative image of the conical pattern of the atomized droplets. Source: Ramisetty et al[49] [Colour figure can be viewed at wileyonlinelibrary.com]

example, air humidification. The obvious variations in local air pressure were linked with the sound waves that occurred all around the resonator and remove the threat of dust and foreign particles accumulation in the locality of the nozzle orifice, therefore eliminating any decay in the atomizer performance. Such a system is dependable, and almost no maintenance is needed in such a case.[51]

Tanyerİ[52] also investigated ultrasound-assisted fuel atomization for internal combustion engine. The ultrasonic sound generation by using field and properties, especially by cavitation effect on liquids, was used. Ultrasonic atomization method was compared with the existing experimental injector atomization data. Scaled droplets occurred as a result of atomization. Image processing method was used to determine the droplet size. The study results showed that ultrasound-assisted atomization enhanced the entire atomization process as compared with the conventional method.

In case of industrial use, ultrasonic atomizers are known to be the finest sprays as they come with a narrow angle full cone jet. These are the best, in a way, as the mixing of water and air is avoided in a confined volume before the particles leave the nozzle, and thus, their feed pressures can be regulated independently without affecting each other. Such a setting permits a wide regulation range on the liquid volume and facilitates the flow of desired operating circumstances.[53] Figure 6 shows an atomizing head gathered onto one A-type adapter, while the B1 AISI 303 stainless steel is used for making material atomizing heads and is also used to make the adapter.

According to previous studies, ultrasonic-assisted atomization showed the feasibility of the application in gas turbines. This technique produces the finest atomized liquid sprays as a full cone spray with a narrow spray angle. Ultrasonic-assisted atomization produced better results compared with the other atomization methods and led to satisfactory operating conditions along a very wide range of fluid capacities.

6 | SUPERCRITICAL FLUID–ASSISTED ATOMIZATION

At supercritical conditions fluid (SCF) can take benefit of certain properties of gases: a constant adaptable solvent/selectivity acquired varying pressure and temperature; diffusivities of two orders of magnitude larger, as compared with those of liquids, are also available. Resultantly, SCF can exhibit quick mass transfer and performances. It is important to note here that this result cannot be obtained from conventional solvents. CO_2 is mostly used if we focus on all potential SCF. Here are the top reasons for this[54]:

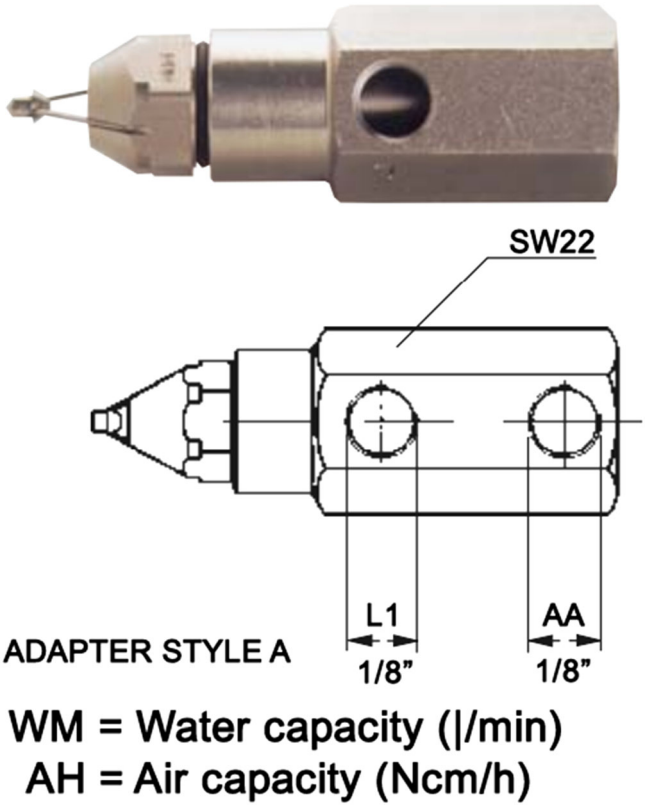

FIGURE 6 Atomizing head assembled onto one A-type adapt [Colour figure can be viewed at wileyonlinelibrary.com]

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- It functions as a lipophilic solvent.
- It is nontoxic, nonflammable, and quite cost-effective.
- Its critical parameters are freely accessible on the industrial scale (T_c = 37.1°C; p_c = 73.8 bar).

The most studied supercritical-based micronization techniques include (1) rapid expansion of supercritical solutions,[55,56] (2) particle generation from gas saturated solutions,[57] (3) supercritical antisolvent precipitation,[58] and (4) supercritical CO_2–assisted atomization.[59]

Therefore, in recent times, SCF is being utilized to develop enhanced and adaptable micronization processes. Supercritical dissolved-gas atomization is basically an atomization process in which CO_2 at temperature and pressure ranging above its critical point is utilized as the atomizing gas. Caputo et al[60] analysed the spray characteristics while keeping in view the droplet size and distribution by using a laser diffraction method that is based on the Malvern apparatus (Figure 7). The droplet size is the gas-to-liquid mass ratio, which is the major characteristic affected by the injection pressure in the range of 7.4 to 13 MPa, has a negligible effect. The droplet mean diameter changes from about 8.0 to 2.0 μm when the value of gas-to-liquid mass ratio ranges from 0.5 to 3; very thin droplet size distributions are also created. The mean droplet diameter is majorly affected by the unexpected release of the gas dissolved in the liquid if the viewpoint of atomization mechanism is taken into account. It is approved from the experimental data that dissolved gas atomization permits the production of micrometric droplets that can result into precipitates with measured sizes and distributions that are beneficial in several fine-particle production methods.

The following four different supercritical gas-assisted atomizer systems were investigated by Gunther et al,[61] and they further compared the atomization performance for water and polymer solution feeds: external mixing, effervescent atomization, superheated atomization, and CO_2-assisted atomization. An external mixing twin-fluid atomizer was used as standard atomization configuration and is compared with three rather unusual atomization techniques. An effervescent atomizer is basically utilized to represent internal mixing atomization system, and it can function at lower air-to-liquid ratio as compared with the conventional gas atomizers. To improve the atomization of liquids by a phase change in the developing spray, superheated atomization is an advanced technique. CO_2-assisted atomization follows the same path by releasing a liquid fully filled with CO_2 into a lower pressure setting. The model fluid water was broken down, and more complex polymer solution with higher viscosities was evaluated with the help of atomization configurations. Resulting spray configurations and drop size distributions were also matched and compared. The following were research results:

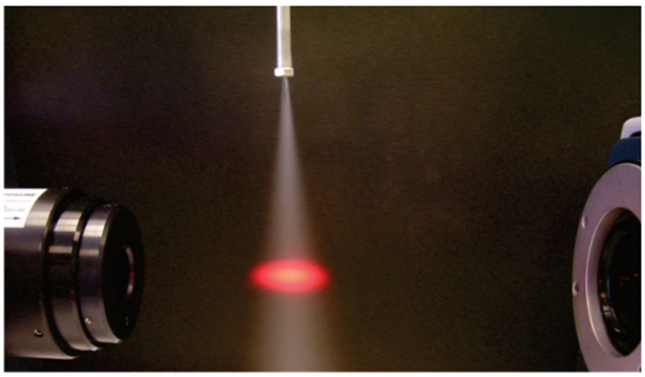

FIGURE 7 Typical spray during droplet analysis by laser diffraction with a laser beam that intersects the spray 7.5-cm downstream of the exit orifice. Source: Caputo et al[60] [Colour figure can be viewed at wileyonlinelibrary.com]

- The droplet sizes formed with CO_2-assisted spraying were larger due to a freezing effect.
- The droplet size distributions of the sprays produced with external, effervescent, or superheated spraying depict that smaller droplet when applied atomization processes in ambient conditions can be achieved.
- Conversely, for higher liquid viscosity, the droplet sizes produced with external and effervescent atomization increased, whereas the droplet sizes measured for superheated atomization remained more or less the same. Additionally, more amounts can be received by using the latter methodology.

Although this novel atomization approach was applied in different gases and compounds, it was not fully applied in gas turbines. In addition, supercritical method of atomization has been shown to produce well-atomized sprays with only a small amount of aerating gas at relatively low injection pressure. Despite the advantages associated with the application of supercritical fluid technique in atomization, there is little or no report on its application in atomizing fuel during gas turbine operations. Therefore, this gap is a baseline for future studies with interest in the use of novel techniques to atomize fuels in gas turbines.

7 | CHALLENGES AND FUTURE DIRECTION

All studies reviewed in the literature support the notion that fuel atomization techniques in gas turbines can be used to enhance more combustion. These techniques increase the fuel combustion efficiency and then decrease the smoke inside the chambers, enhancing the turbine

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TABLE 4 Comparison of atomization techniques

	Key Characteristics/ Requirements to Achieve High Efficiency	Advantages	Disadvantages	Limitations	Other Aspect
Air-assisted atomization	The optimization of the droplets resulted from this method depends on the atomized sprays.	✓ Low volume ✓ High pressure ✓ Provide good atomization of fuels ✓ Create high particle velocity ✓ Produce fine droplet	✓ Utilizes more compressed air for the atomization process ✓ High-cost pressure pumps	✓ The air should be filtered at system inlet due to the narrow inside passages. ✓ Air-assisted atomization possible to obtain jets with limited spray angles due to the high speed of the spray.	✓The efficiency can be reached up to 65%. ✓ Spray pattern often present problems. ✓ Wide range of atomized droplet distribution in spray pattern. ✓ Droplet size ranging from 17 to 200 μm
Plasma-assisted atomization	The optimization of this technique depends on the electrical discharges on complicated chemical and kinetic processes within the combustors.	✓ Produce high temperature, which results in high atomization efficiency ✓ Less chemical interference ✓ Atomization in the inter (argon) atmosphere	It is complicated	Having controllable and repeatable spray flow for a wide range of operating conditions and various particle concentrations of spray systems	✓ The droplet size ranges from 1 to 200 μm. ✓ Can approach temperature up to 9726.85°C
Ultrasonic-assisted atomization	For optimum atomization, factors such as viscosity, miscibility, and solid content should be considered. For example, the viscosity should be under 500 cps and the solid concentration kept below 30%	✓ Produce very fine sprays ✓ Low-velocity spray ✓ Smaller drop sizes, which can be accurately controlled ✓ More uniform size distribution ✓ High sphericity of droplet ✓ Prevent dust and lime particles from building up at the orifice and impairing atomizer performance.	The sound waves generate a typical noise	Noise restricts their use to such systems where personnel is not exposed to noise levels exceeding the limit values specified by local safety regulations	✓ The vibrations with a frequency in excess of 20 kHz. ✓ Droplet sizes are ranging from 1 to 100 μm.
Supercritical fluid–assisted atomization	The optimum of the SCF method depends on the optimum values set of temperature and pressure.	✓ Speed ✓ Selectivity ✓ Environmental improvement ✓ Reduced product contamination	High cost due to the high pressures needed	CO_2 itself is non-polar, and limited dissolving power, so cannot always be used as a solvent on its own, particularly for polar solutes	The droplet sizes are arranging from 2.0 to 8.0 μm.

Abbreviation: SCF, supercritical conditions fluid.

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efficiency. However, there are several limitations on the wide acceptance and commercialization of fuel atomization techniques in gas turbines, for example, air-assisted atomization. One of the challenges included the have proper filtration at system inlet and obtain jets with limited spray angles. These challenges happened because of the narrow internal passages and the high-speed spray. Therefore, there is a need to identify suitable approaches to overcome these challenges. One of the ways that can be applied to overcome is to design and have multiple orifice air nozzles. Sound waves play a very important role by using ultrasound-assisted atomization technique in gas turbines. This is important because the sound waves will identify the required droplet size. In addition, the principle of ultrasound-assisted atomization operation demands sound waves to be produced and the consequent operational noise restricts their use to such systems where personnel is not exposed to noise levels exceeding the limit values specified by local safety regulations. Al-Jumaily and Meshkinzar[62] stated that the issues lying behind ultrasound generation in gases are attributed to low specific acoustic impedance and high absorption of the medium. Consequently, this limitation is overcome by having good impedance matching between the transducer and gas medium, large amplitudes of vibration, and highly directional radiation were needed.

Another major challenge for fuel atomization techniques in gas turbines is associated with plasma method. In this technique, the spray systems are highly concerned to have a controllable and repeatable spray flow for a wide range of operating conditions and various particle concentrations. However, this challenge can be overcome by creating innovative designs for the plasma atomization in the gas turbines. For this, the spray atomization can be improved by having a coannular flow inside the discharge passage.

More so, the process of CO_2 in the supercritical method is another challenge for atomization techniques. CO_2 itself is a nonpolar and has limited dissolving power, so it cannot be used as always a solvent on its own, particularly for polar solutes. This can affect the efficiency of using this technique in gas turbines. Therefore, to overcome this limitation, it is feasible to study and use another solvent than CO_2, which can be used as a solvent on its own. Therefore, more study in this regard is required. The overall comparisons of four major techniques are summarized in Table 4.

8 | CONCLUSION

The main goal of the current paper is to review and summarize the application of a novel technology in fuel atomization. With respect to gas turbines, the four major categories investigated were the (1) air-assisted, (2) plasma-assisted, (3) ultrasonic-assisted, and (4) supercritical fluid–assisted. This study found that the different novel techniques discussed have potentials in effective fuel atomization.

The scientific literatures reviewed have shown that the application of atomization techniques depend on the optimization results, such as the droplet size. If all techniques were taken into account, apart from the equipment operating parameters, droplet size was also thought to be dependent on the liquid phase physicochemical properties like surface tension and viscosity or on the rheological behaviour. In addition, the droplet size distribution was reliant on the flow rate of the liquid, and it is important to note here that the distribution increases as the liquid flow rate increases and two peaks were recorded linked to the formation of secondary droplets. The air-assisted atomization produced droplet size that ranged between 17 and 200 μm, and the plasma-assisted atomization had produced droplet size of 1 to 200 μm. Meanwhile, the droplet size produced ultrasound-assisted atomization was from 1 to 100 μm and the droplet size produced by supercritical fluid–assisted atomization was from 2 to 8 μm. Therefore, plasma- and ultrasound-assisted techniques were found to be the best as compared with the others due to their production of the finest droplet size. In terms of cost in adopting these technologies, the ultrasound-assisted and supercritical fluid–assisted techniques were the least expensive to set up and easy to adapt as compared with other techniques (air- and plasma-assisted techniques).

In conclusion, this review suggests that the desired droplet size distribution with the required mean size can be attained if the equipment operating characteristics (power dissipation, frequency, and flow rate) was fully controlled along with the liquid physicochemical possessions (viscosity, sort of the liquid, and surface tension). Although they are generally limited by (1) the proper filtration at system inlet and obtained jets with limited spray angles, (2) noise, (3) controllable and repeatable spray flow for a wide range of operating conditions and various particle concentrations, and (4) the limitation of CO_2 cannot be used always as a solvent on its own. Therefore, more studies should focus on enhancing their acceptability and efficiency.

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